3. In this question you must show all stages of your working.

Solutions relying entirely on calculator technology are not acceptable.

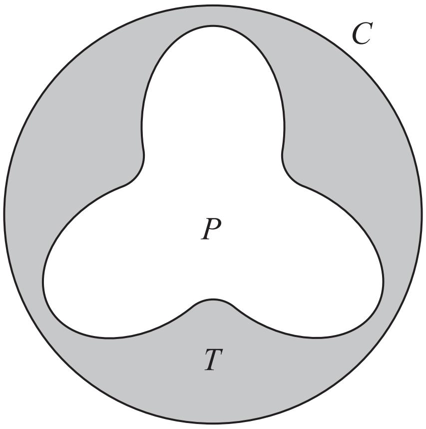

Figure 1

Figure 1 shows the design for a bathing pool.

The pool, P, shown unshaded in Figure 1, is surrounded by a tiled area, T, shown shaded in Figure 1.

The tiled area is bounded by the edge of the pool and by a circle, C, with radius 6 m.

The centre of the pool and the centre of the circle are the same point.

The edge of the pool is modelled by the curve with polar equation

$$r = 4 - a\sin 3\theta \qquad 0 \leqslant \theta \leqslant 2\pi$$

where a is a positive constant.

Given that the shortest distance between the edge of the pool and the circle C is 0.5 m,

(a) determine the value of a.

(2)

(b) Hence, using algebraic integration, determine, according to the model, the exact area of T.

(6)

3.

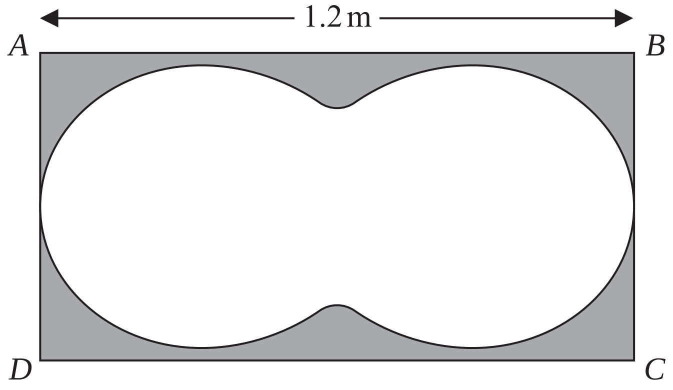

Diagram not to scale

Figure 1

Figure 1 shows the design for a table top in the shape of a rectangle *ABCD*. The length of the table, *AB*, is 1.2 m. The area inside the closed curve is made of glass and the surrounding area, shown shaded in Figure 1, is made of wood.

The perimeter of the glass is modelled by the curve with polar equation

$$r = 0.4 + a\cos 2\theta \qquad 0 \leqslant \theta < 2\pi$$

where a is a constant.

(a) Show that $a = 0.2$

(2)

Hence, given that $AD = 60$ cm,

(b) find the area of the wooden part of the table top, giving your answer in m^2 to 3 significant figures.

(8)

3.

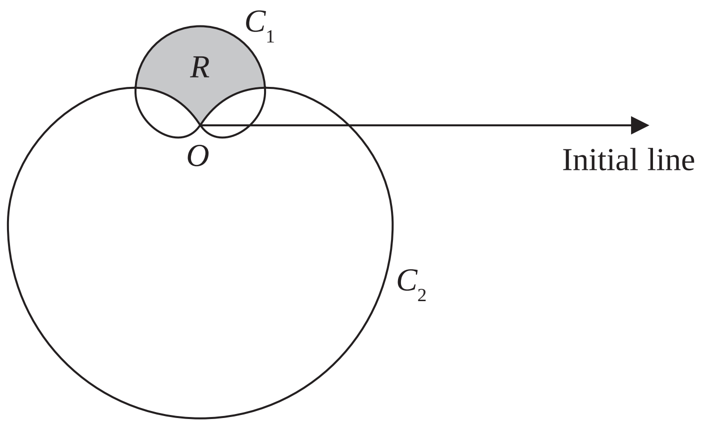

Figure 1

Figure 1 shows a sketch of two curves C_1 and C_2 with polar equations

$$C_1: r = (1 + \sin\theta) \qquad 0 \leqslant \theta < 2\pi$$

$$C_2: r = 3(1 - \sin\theta) \qquad 0 \leqslant \theta < 2\pi$$

The region R lies inside C_1 and outside C_2 and is shown shaded in Figure 1.

Show that the area of R is

$$p\sqrt{3} - q\pi$$

where p and q are integers to be determined.

(9)

5. The locus C is given by

$$|z-4| = 4$$

The locus D is given by

$$\arg z = \frac{\pi}{3}$$

(a) Sketch, on the same Argand diagram, the locus C and the locus D

(4)

The set of points A is defined by

$$A = \{z \in \mathbb{C} : |z-4| \leqslant 4\} \cap \left\{z \in \mathbb{C} : 0 \leqslant \arg z \leqslant \frac{\pi}{3}\right\}$$

(b) Show, by shading on your Argand diagram, the set of points A

(1)

(c) Find the area of the region defined by A, giving your answer in the form $p\pi + q\sqrt{3}$ where p and q are constants to be determined.

(4)

P 7 5 6 8 3 A 0 1 2 3 2

9.

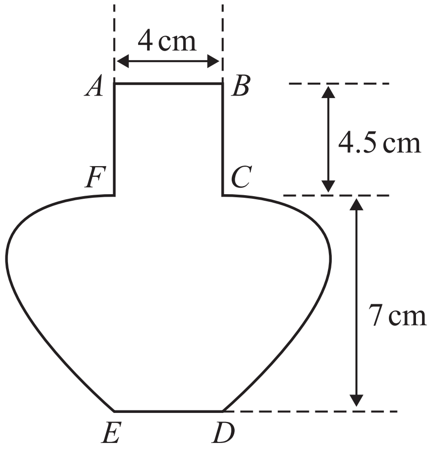

Figure 1

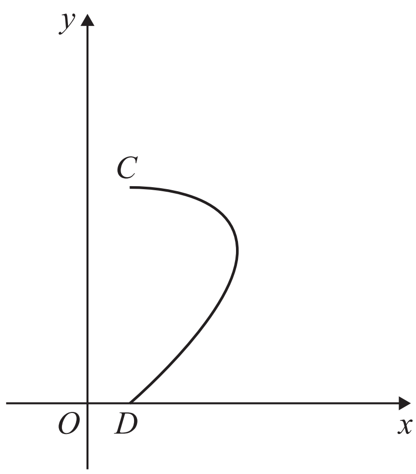

Figure 2

Figure 1 shows the central vertical cross-section *ABCDEFA* of a vase together with measurements that have been taken from the vase.

The horizontal cross-section between *AB* and *FC* is a circle with diameter 4 cm.

The base of the vase *ED* is horizontal and the point *E* is vertically below *F* and the point *D* is vertically below *C*.

Using these measurements, the curve *CD* is modelled by the parametric equations

$$x = a + 3\sin 2t \qquad y = b\cos t \qquad 0 \leqslant t \leqslant \frac{\pi}{2}$$

where *a* and *b* are constants and *O* is the fixed origin, as shown in Figure 2.

(a) Determine the value of *a* and the value of *b* according to the model.

(2)

(b) Using algebraic integration and showing all your working, determine, according to the model, the volume of the vase, giving your answer to the nearest cm^3

(7)

(c) State a limitation of the model.

(1)

6. The curve C has equation

$$r = a(p + 2\cos\theta) \qquad 0 \leqslant \theta < 2\pi$$

where a and p are positive constants and $p > 2$

There are exactly four points on C where the tangent is perpendicular to the initial line.

(a) Show that the range of possible values for p is

$$2 < p < 4$$

(5)

(b) Sketch the curve with equation

$$r = a(3 + 2\cos\theta) \qquad 0 \leqslant \theta < 2\pi \qquad \text{where } a > 0$$

(1)

John digs a hole in his garden in order to make a pond.

The pond has a uniform horizontal cross section that is modelled by the curve with equation

$$r = 20(3 + 2\cos\theta) \qquad 0 \leqslant \theta < 2\pi$$

where r is measured in centimetres.

The depth of the pond is 90 centimetres.

Water flows through a hosepipe into the pond at a rate of 50 litres per minute.

Given that the pond is initially empty,

(c) determine how long it will take to completely fill the pond with water using the hosepipe, according to the model. Give your answer to the nearest minute.

(7)

(d) State a limitation of the model.

(1)

7.

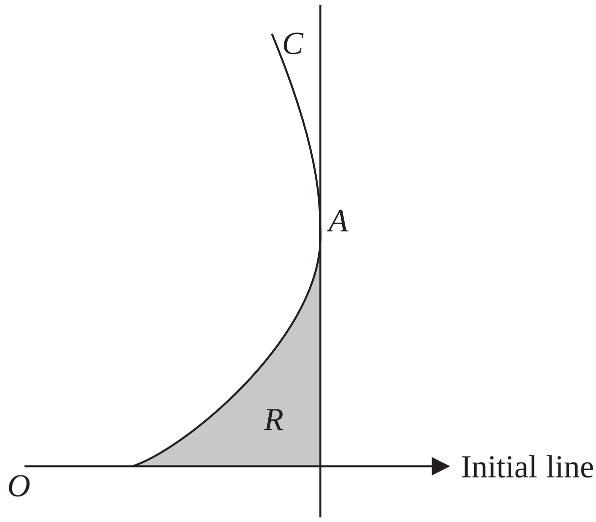

Figure 1

Figure 1 shows a sketch of the curve C with equation

$$r = 1 + \tan\theta \qquad\qquad 0 \leqslant \theta < \frac{\pi}{3}$$

Figure 1 also shows the tangent to C at the point A.
This tangent is perpendicular to the initial line.

(a) Use differentiation to prove that the polar coordinates of A are $\left(2, \frac{\pi}{4}\right)$

(4)

The finite region R, shown shaded in Figure 1, is bounded by C, the tangent at A and the initial line.

(b) Use calculus to show that the exact area of R is $\frac{1}{2}(1 - \ln 2)$

(6)

1.

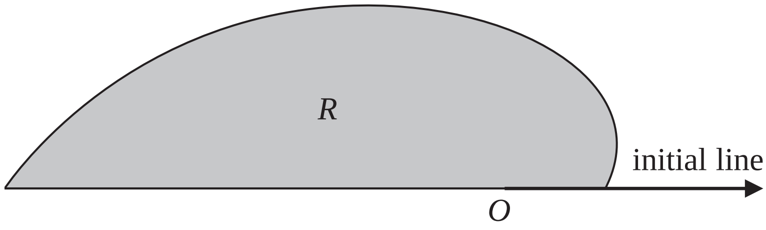

Figure 1

Figure 1 shows a sketch of the curve with polar equation

$$r = 2\sqrt{\sinh\theta + \cosh\theta} \qquad 0 \leqslant \theta \leqslant \pi$$

The region R, shown shaded in Figure 1, is bounded by the initial line, the curve and the line with equation $\theta = \pi$

Use algebraic integration to determine the exact area of R giving your answer in the form $pe^q - r$ where p, q and r are real numbers to be found.

(4)

4. (a) Sketch the polar curve C, with equation

$$r = 3 + \sqrt{5}\cos\theta \qquad 0 \leqslant \theta \leqslant 2\pi$$

On your sketch clearly label the pole, the initial line and the value of r at the point where the curve intersects the initial line.

(2)

The tangent to C at the point A, where $0 < \theta < \frac{\pi}{2}$, is parallel to the initial line.

(b) Use calculus to show that at A

$$\cos\theta = \frac{1}{\sqrt{5}}$$

(4)

(c) Hence determine the value of r at A.

(1)